INCIDENT COMMUNICATION GENERATOR

Automated, Accurate and Audience-Specific Cybersecurity Incident Updates

ORGANISATION

RedKross Research Foundation

DOMAIN

SOC Operation and Threat Detection

TASK ID

2438

The Communication Challenge

- Security incidents require timely and accurate communication.
- Different stakeholders need different levels of technical detail.
- Manual communication can lead to inconsistent messages.
- Unverified information may be mistaken for confirmed facts.
- Missing update schedules and contact information can create confusion.

01

Information inconsistency

Manual drafting produces differing messages.

02

Audience communication gaps

Each group needs a different level of detail.

03

Unverified assumptions

Hypotheses can be mistaken for confirmed facts.

04

Delayed updates

Missing update times and contact routes cause confusion.

Our Solution

Transforms structured incident information into audience-specific communication reports.

01 Structured incident input

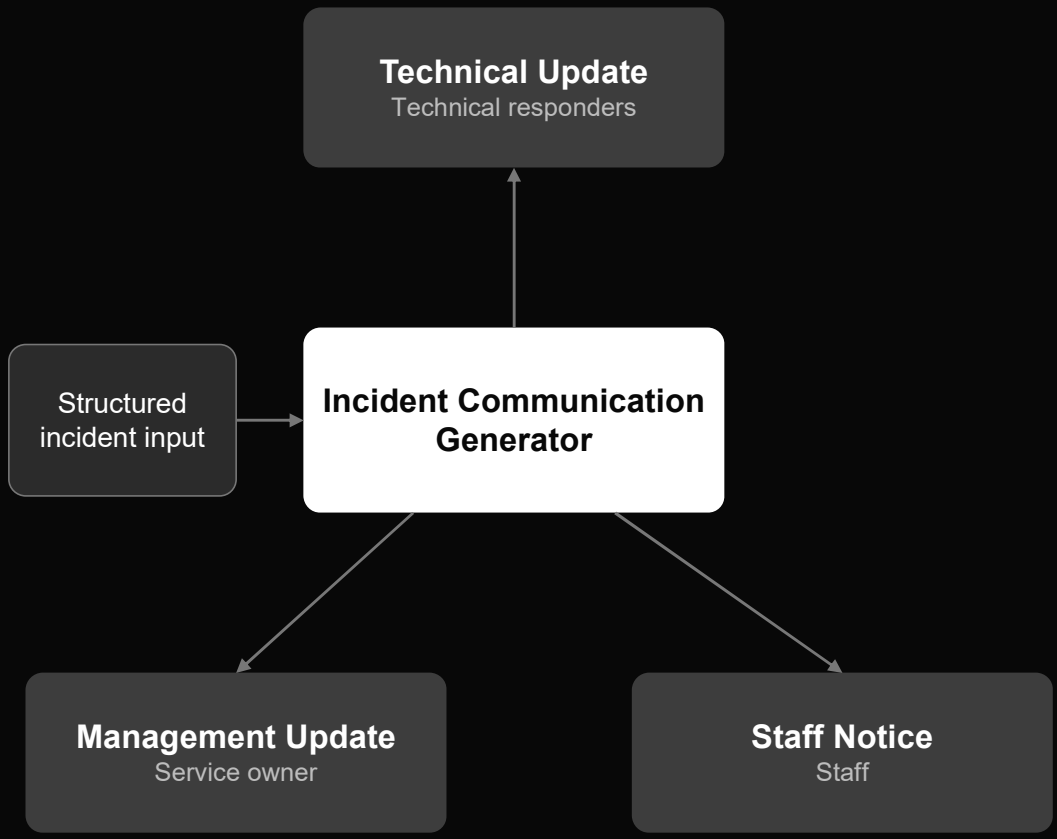
02 Schema-based input validation

03 Automated report generation

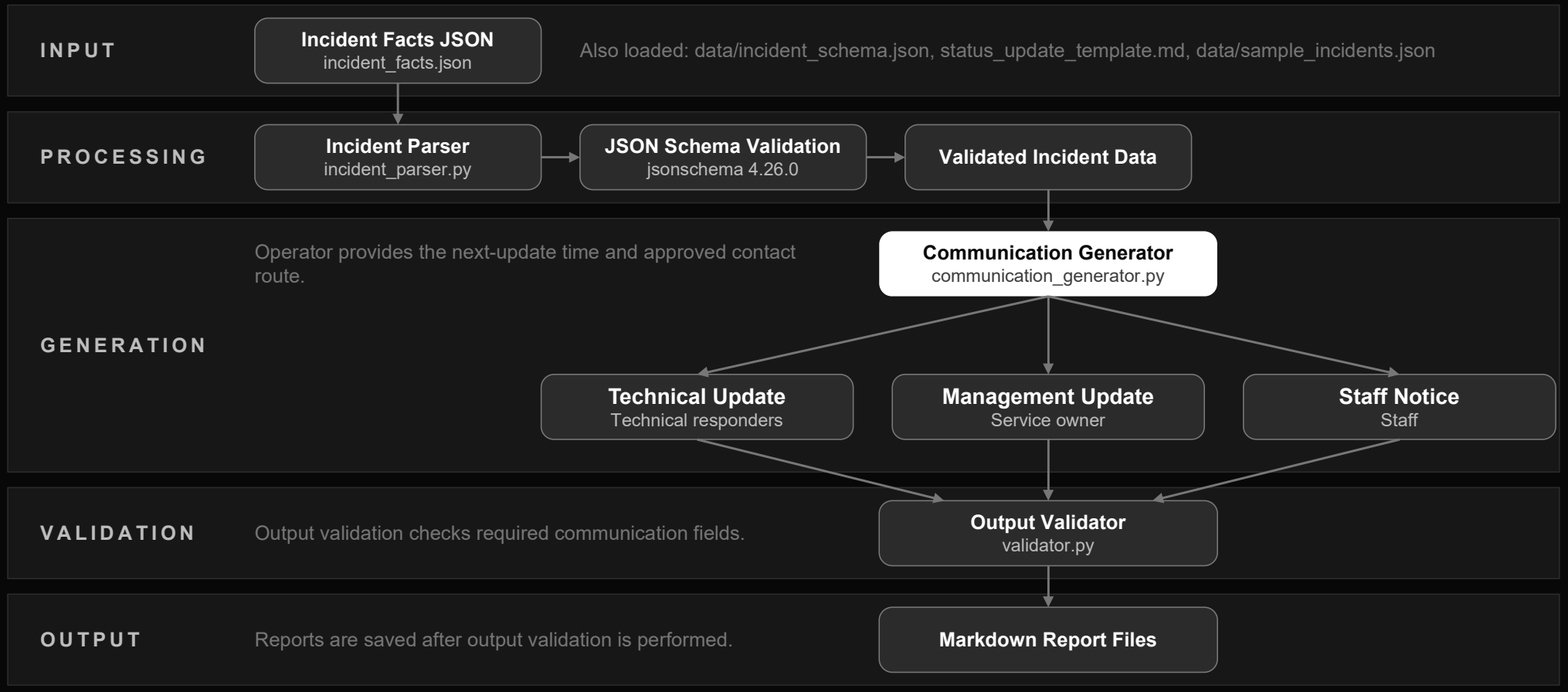
04 Audience-specific content

05 Output validation

06 Markdown report generation



System Architecture



Technology Behind the System

LANGUAGE

Python 3.12

Core application development.

DATA FORMAT

JSON

Structured incident information.

VALIDATION

JSON Schema

Input structure validation.

LIBRARY

jsonschema 4.26.0

Schema validation library.

OUTPUT FORMAT

Markdown

Generated communication reports.

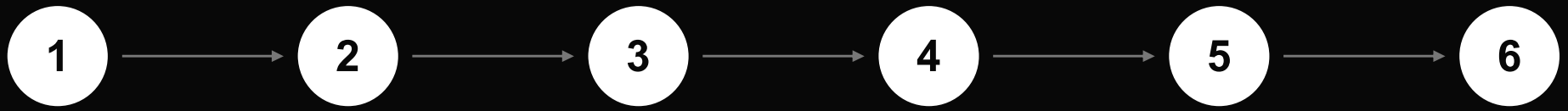
TESTING

Python unittest

Automated testing.

CORE MODULES incident_parser.py · communication_generator.py · validator.py · main.py

From Incident Data to Communication



STAGE 01

Load Data

Incident data is loaded.

STAGE 02

Validate Schema

JSON structure is validated against the schema.

STAGE 03

Next-Update Time

The operator provides the next update time.

STAGE 04

Contact Route

The operator provides the approved contact route.

STAGE 05

Generate

Communication is generated for three audiences.

STAGE 06

Validate Output

Output validation is performed before reports are saved.

SYSTEM — OPERATOR-PROVIDED — SYSTEM

One Incident. Three Audiences.

TECHNICAL RESPONDERS

Technical Incident Update

- Confirmed authentication events.
- Unresolved investigation points.
- Recommended technical response actions.
- Incident status and update information.

SERVICE OWNER

Management Incident Update

- Current incident situation.
- Confirmed and unconfirmed information.
- Business impact currently known.
- Recommended response actions.

STAFF

Staff Security Notice

- What is currently known.
- What is not yet confirmed.
- Security precautions.
- Approved reporting contact route.

Understanding the Security Incident

Incident ID: RK-INC-TRAIN-204 | **Northstar Community Services** | **Status: Investigating**

CONFIRMED FACTS

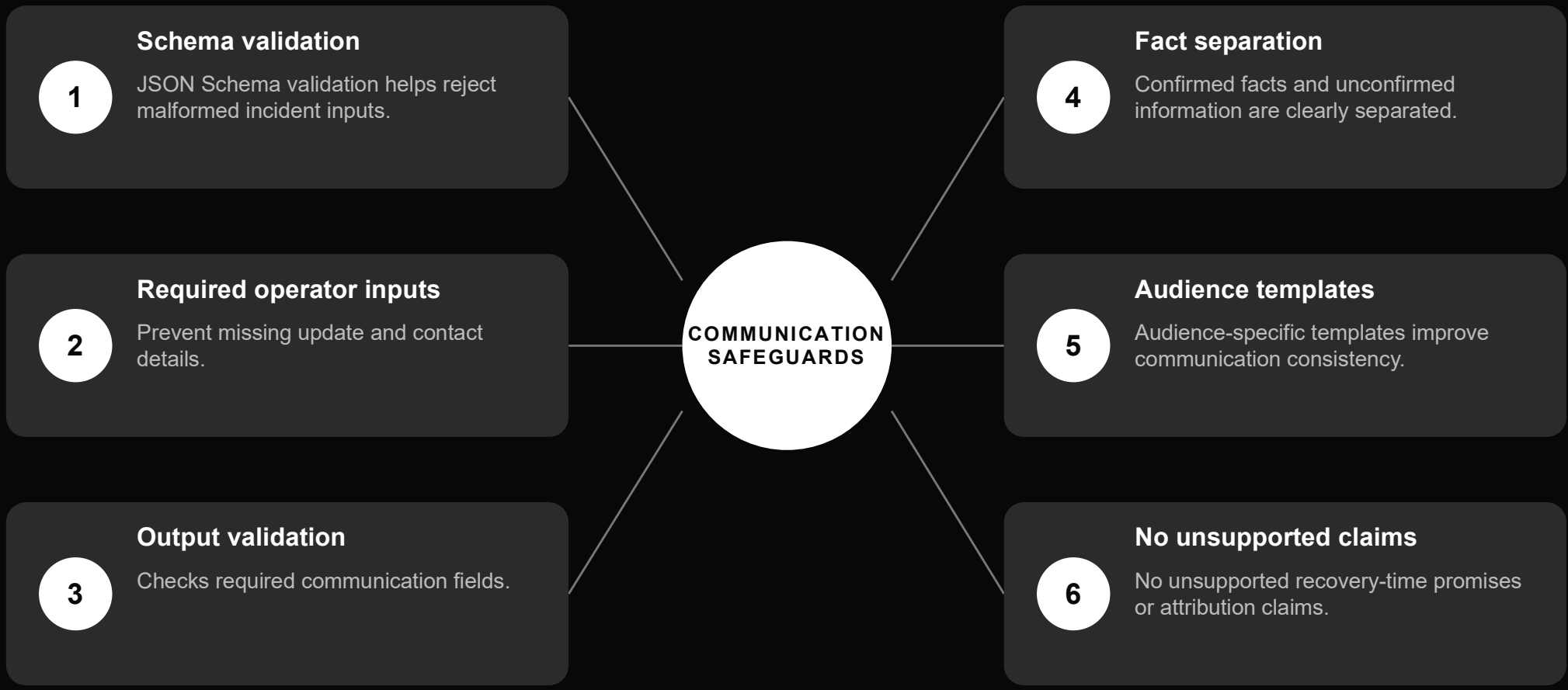
- Repeated failed sign-in events.
- Two unexpected MFA prompts denied by the user.
- Privilege event with unclear change-ticket reference.

UNDER INVESTIGATION

- Possible credential theft.
- Possible data access or copying.
- Whether the privilege event was unauthorised.

Hypotheses are listed separately and are not presented as established facts.

Built with Communication Safeguards



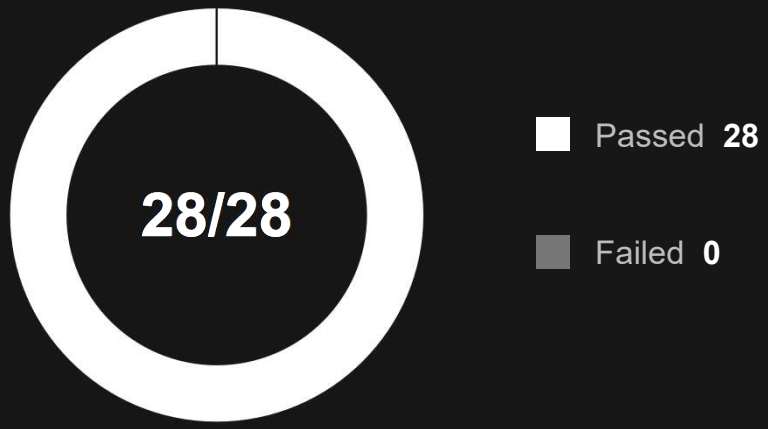
Verified Through Automated Testing

28	TOTAL TESTS

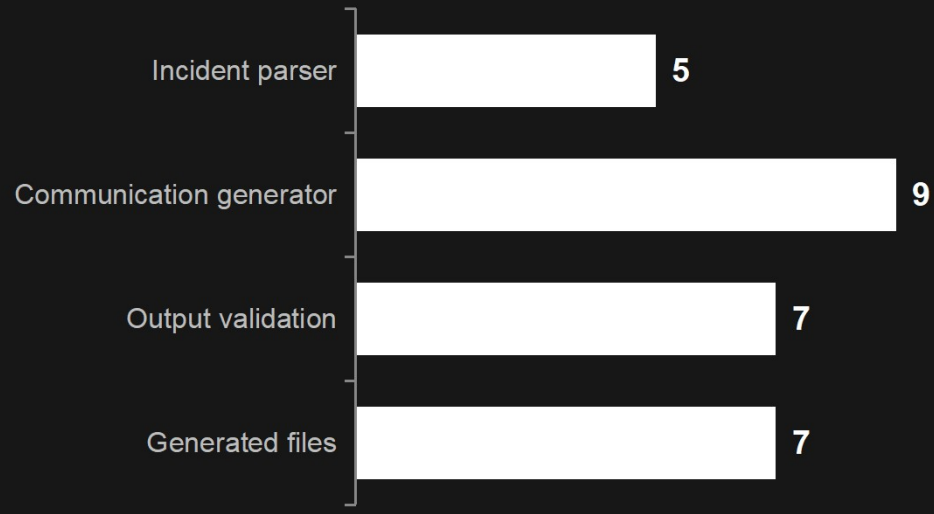
28	PASSED

0	FAILED

TEST RESULTS



TESTS BY CATEGORY



Results and Future Direction

IMPLEMENTED

PROJECT OUTCOMES

- Automated generation of three communication formats.
- Consistent use of structured incident information.
- Clear separation of confirmed and unconfirmed facts.
- Validated Markdown report outputs.
- 28 successful automated tests.

CURRENT LIMITATIONS

- Uses structured input rather than live security-platform integrations.
- Requires an operator to provide update time and contact route.
- Output validation checks structure and required indicators, not full semantic truth.
- Human review is still necessary before distributing incident communications.

NOT YET IMPLEMENTED

FUTURE ROADMAP

1. SOC platform integration.
2. More configurable communication templates.
3. Role-based approval workflow.
4. Audit logging.
5. Enhanced semantic consistency checks.

THANK YOU

Building Clearer, Safer and More Consistent Incident Communication.

PROJECT

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